

Clinical Budgeting and Project Management System

Prototype User Guide

A drill-down clinical study planning prototype where a budget line becomes an expandable work unit with composition, schedule, location, evidence, payment logic, cost, resources, documents, and earned value tracking.

Illustration only. The demo uses public protocol references and simplified assumptions. Budget numbers, contract text, monitoring language, PM plan language, SOP placeholders, and schedule outputs are not correct for contracting, regulatory use, quality system use, or operational decision-making.

1. Purpose Of The System

The system demonstrates a planning approach for clinical research where the budget is not a flat spreadsheet. Each budget line is a unit of work that can be planned at a high level first, then progressively detailed as information becomes available.

It starts with protocol-derived units, study phases, milestones, global counts, role-based labor, pass-through costs, and investigator grants.

It becomes a country/site schedule, resource plan, timesheet plan, evidence tracker, payment tracker, change order register, and earned value dashboard.

2. Budgeting Process

1. Extract units from the protocol: visits, start-up, monitoring, safety, data, PM, reporting, close-out, and follow-up activities.
2. Create budget units with phase, milestone, count, location assumption, dates, and cost model.
3. Define role composition: department, role, function, planned hours per unit, hourly rate, and optional employee assignment.
4. Split detail progressively from Global to Region to Country to Site.
5. Generate schedule dates and override exact planned dates when known.
6. Track completion, evidence, actual hours, validation, and payment status.
7. Recalculate budget, resources, cash flow, timesheets, evidence, EVA, and documents from the same unit model.

3. Core Data Model

Budget Unit: main expandable work unit, for example Monitoring Visit or Site Start-Up Package.

Unit Composition: role rows used to calculate labor cost and FTE.

Unit Schedule: generated or manually edited unit instances.

Completion: actual completion date, completed by, and actual hours.

Evidence Document: proof of completion and validation, for example Trip Report, CAPA, SAE, or Other.

Payment Status: investigator payment and sponsor reimbursement tracking.

Commercial Source: Contract Unit, CO1, CO2, OOS1, OOS2, etc.

4. Dashboard And KPIs

The top of the app shows the project, protocol source, disclaimer, timeline, Budget Story, KPI cards, and progress indicators.

KPI cards include Planned Budget, Labor Budget, Planned Hours, FTE Effort, Average FTE Need, Pass-through, Investigator Grants, and Actual Cost.

Progress indicators show Scheduled, Completed, Evidence Filed, and Paid by Sponsor.

5. Budget Grid

The Budget Grid is the commercial and operational starting point. One row equals one expandable work unit.

Columns include Unit list, Phase, Milestone, Commercial source, Location, Number of units, Pass-through per unit, Grant per unit, Start, End, Status, and Actions.

Commercial source defaults to Contract Unit and can be changed to CO1, CO2, CO3, OOS1, OOS2, or OOS3.

Actions include Split country, Save, Open, and Delete.

6. Unit Drill-Down

Each saved row opens as a unit card with tabs: Detail Builder, Unit Composition, Duration & Schedule, Completion & Evidence, Payment Logic, and Cost Summary.

Detail Builder shows maturity ladders for Location, Staffing, Timing, and Actuals.

Unit Composition allows editing department, job role, function, employee, hours per unit, hourly rate, milestone, and assumption type.

Duration & Schedule generates dates between start and end dates and allows exact date overrides.

Completion & Evidence records actual completion, actual hours, evidence status, and proof link placeholders.

Payment Logic tracks investigator payment, sponsor reimbursement, and payment trigger.

Cost Summary calculates planned cost, actual cost, variance, validated cost, not validated cost, and open exposure.

7. Report Tabs

Plan: budget grid and expandable unit cards.

Budget: unit, source, phase, milestone, units, hours, FTE-months, labor, pass-through, grant, and total.

Resources: monthly resource demand by department, role, employee, hours, and FTE.

Cash Flow: monthly labor, pass-through, investigator grant, and total cash flow.

Timesheets: generated planned timesheet rows and actual-hour comparison.

Evidence: document status and payment evidence tracker.

EVA: Planned Value, Earned Value, Actual Cost, SPI, CPI, and unit-level EVA.

Contracts & Docs: contract backbone, commercial summary, change order/OOS register, PM plan, monitoring plan, and SOP/TMF placeholders.

8. Contract, Change Order, And OOS Logic

Contract Unit means baseline scope included in the original contract or work order.

CO1, CO2, and CO3 mean approved or planned change order scope.

OOS1, OOS2, and OOS3 mean out-of-scope work not yet converted to a change order.

This allows the same operational budget to support contract control, budget control, and project reporting.

9. Document Constructor

The Contracts & Docs tab uses the live budget grid to summarize commercial scope and create document backbones.

It includes a Study Work Order / Clinical Services Agreement outline between Muster Sponsor AG and Muster CRO GmbH.

It also includes a Commercial Summary, Change Order / OOS Register, Project Management Plan backbone, Monitoring Plan backbone, and SOP/TMF placeholder list.

10. Recommended Demo Workflow

1. Start on the Plan tab and review Budget Story and KPIs.
2. Change one budget row from Contract Unit to OOS1, then Save.
3. Use Country split to show how global units become country, site, and TBD remainder.
4. Open a unit card and show the Detail Builder maturity ladder.
5. Edit a planned date in Duration & Schedule.
6. Mark an item completed and change evidence from Draft to Filed.
7. Show that Budget, Resources, Cash Flow, Evidence, EVA, and Contracts & Docs update from the same model.

11. Current Prototype Limits

No backend, database, login, audit trail, validation package, role security, or electronic signature.

No real document storage. Document links are placeholders.

No real protocol parsing pipeline. The current demo uses simplified manual assumptions.

No real legal contract generation. Contract language is sample structure only.

No regulatory-ready SOPs. SOP names and sections are placeholders for product concept discussion.

No persistent save after browser refresh unless extended with local storage or backend

storage.